

ADS QUIZ-6

Total points 19/20 ?

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✓ 1. What is the purpose of a hash function in hash mapping? *

1/1

- ☒ To convert the key into an index in the hash table.
- ☐ To encrypt the keys to ensure security.
- ☐ To ensure that each key is unique.
- ☐ To compress the key space to a smaller range of values.



✓ 2. Which of the following is a valid hash function? *

1/1

- ☒ $h(x) = x \% 10$
- ☐ $h(x) = x * 2$
- ☐ $h(x) = x / 5$
- ☐ $h(x) = x + 5$



✓ 3. Which of the following data structures is typically used for collision resolution in hash tables?

*1/1

- ☐ Stacks
- ☐ Arrays
- ☒ Linked lists
- ☐ Queues



✓ 4. _____ is a collision resolution technique where the hash function is repeatedly applied to the next index until an empty slot is found

*1/1

- ☐ Quadratic probing
- ☒ Linear probing
- ☐ Separate chaining
- ☐ Coalesced hashing



✓ 5. When hashing function generates same value for multiple different keys then it is known as _____. *1/1

- ☐ Collision resolution
- ☐ Collision function
- ☐ Multiple Hashing
- ☒ Hash Collision



✓ 6. Which of the following is a collision resolution method in hash tables that uses separate linked lists to store collided elements? *1/1

- ☐ Linear probing
- ☒ Chaining
- ☐ Double hashing
- ☐ Quadratic probing



✓ 7. Which of the following is a disadvantage of using linear probing as a collision resolution method in hash tables? *1/1

- ☒ It may cause clustering.
- ☐ It may cause long chains.
- ☐ It may require a large amount of memory.
- ☐ It may not guarantee that all slots in the table are used.



✗ 8. Which of the following is a collision resolution method in hash tables that uses a quadratic function to compute the next probe sequence? *0/1

- ☐ Linear probing
- ☐ Quadratic probing
- ☐ Double hashing
- ☒ Chaining

✗

Correct answer

- ☒ Quadratic probing

✓ 9. Which of the following is a collision resolution method in hash tables that uses a second hash function to compute the next probe sequence? *1/1

- ☐ Linear probing
- ☐ Quadratic probing
- ☒ Double hashing
- ☐ Chaining

✓

✓ 10. Which of the following graphs has no cycles? * 1/1

- ☐ Directed graph
- ☒ Non-directed graph
- ☐ Connected graph
- ☐ Strongly connected graph

✓



✓ 11. What is an articulation point in a non-connected graph? * 1/1

- ☐ A point where two edges intersect
- ☐ A point where an edge terminates
- ☒ A vertex whose removal makes the graph non-connected
- ☐ A vertex whose removal doesn't affect the graph



✓ 12. Which of the following is true about a strongly connected graph? * 1/1

- ☐ Every vertex has the same degree
- ☐ There are no cycles in the graph
- ☒ Every vertex can be reached from every other vertex
- ☐ Every edge has a weight associated with it



✓ 13. What is a path in a graph? * 1/1

- ☐ The set of all vertices in a graph
- ☐ The set of all edges in a graph
- ☐ The set of vertices that have at least one edge connecting them
- ☒ The set of vertices between a pair of vertices



✓ 14. What is a directed acyclic graph (DAG)? *

1/1

- ☒ A graph with no cycles and directed edges
- ☐ A graph with cycles and directed edges
- ☐ A graph with no cycles and non-directed edges
- ☐ A graph with cycles and non-directed edges



✓ 15. What data structure is used in DFS traversal to keep track of visited vertices? *1/1

- ☒ Stack
- ☐ Queue
- ☐ Array
- ☐ Linked list



✓ 16. What is the time complexity of heap sort? *

1/1

- ☐ $O(n)$
- ☒ $O(n \log n)$
- ☐ $O(n^2)$
- ☐ $O(\log n)$



✓ 17. What is the basic idea behind heap sort? *

1/1

- ☐ Partition the input array into smaller subarrays and sort them recursively
- ☐ Swap adjacent elements if they are in the wrong order until the array is sorted
- ☒ Build a heap data structure from the input array and repeatedly remove the root element ✓
- ☐ Use a divide-and-conquer approach to sort the array

✓ 18. Which of the following statements about inserting and removing elements in a max heap is correct? *1/1

- ☐ Adding a new element to a max heap takes $O(1)$ time.
- ☒ Adding a new element to a max heap takes $O(\log n)$ time, where n is the number of elements in the heap. ✓
- ☐ Removing an element from a max heap takes $O(1)$ time.
- ☐ Removing an element from a max heap takes $O(\log n)$ time, where n is the number of elements in the heap.

✓ 19. What is the time complexity of Heap sort in the worst case? *

1/1

- ☒ $O(n \log n)$ ✓
- ☐ $O(n^2)$
- ☐ $O(n)$
- ☐ $O(\log n)$



✓ 20. Which of the following data structures is used to implement Heap sort? *1/1

- ☐ Linked list
- ☐ Queue
- ☐ Stack
- ☒ Binary heap



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